Randomized algorithms
MODULE-6

The Hiring Problem

We will now begin our investigation of randomized algorithms with a toy problem:

- You want to hire an office assistant from an employment agency.
- You want to interview candidates and determine if they are better than the current assistant and if so replace the current assistant.
- You are going to eventually interview every candidate from a pool of n candidates.
- You want to always have the best person for this job, so you will replace an assistant with a better one as soon as you are done the interview.
- However, there is a cost to fire and then hire someone.
- You want to know the expected price of following this strategy until all n candidates have been interviewed.

- Number of candidates = n.
- Interview one candidate per day decide whether to hire the person or not (Cost of interviewing a candidate = α
- If the candidate is better than the current assistant, must fire the assistant and hire the candidate. (Cost of hiring any candidate =ch.)
- **Goal: Estimate the cost of this strategy**

Assumption

- The candidates for the office assistant job are numbered 1 through n.
- creates a dummy candidate, numbered 0, who is less qualified than each of the other candidates.
- after interviewing candidate i , determine if candidate i is the best candidate you have seen so far.

To estimate the cost :

- Since all candidates must be interviewed, the interview cost n is unavoidable.
- If m candidates are hired, the hiring cost is much. This cost varies with the input. So, we focus on this cost

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```
Hire-Assistant(n)
best <- dummy candidate
for i <- 1 to n
    do interview of candidate i
        if candidate i is better than best
            then best <- i
                hire candidate i
```

Total Cost and Cost of Hiring

- Interviewing has a low cost c_i.
- Hiring has a high cost c_h.
- Let n be the number of candidates to be interviewed and let m be the number of people hired.
- The total cost then goes as $O(n^*c_i+m^*c_h)$
- The number of candidates is fixed so the part of the algorithm we want to focus on is the m^*c_h term.
- This term governs the cost of hiring.

Worst-case Analysis

- In the worst case, everyone we interview turns out to be better than the person we currently have.
- In this case, the hiring cost for the algorithm will be $O(n*c_h)$.
- This bad situation presumably doesn't typically happen so it is interesting to ask what happens in the average case.

Randomized algorithms

- In order to use probabilistic analysis, we need to know something about the distribution of the inputs.
- Unfortunately, often little is known about this distribution.
- We can nevertheless use probability and analysis as a tool for algorithm design by having the algorithm we run do some kind of randomization of the inputs.
- This could be done with a random number generator. i.e.,
- We could assume we have primitive function Random(a,b) which returns an integer between integers a and b inclusive with equally likelihood.
- Algorithms which make use of such a generator are called randomized algorithms.
- In our hiring example we could try to use such a generator to create a random permutation of the input and then run the hiring algorithm on that.

Randomized algorithm for Hiring Assistant:

Assumptions:

- The list of candidates is an input to the algorithm.
- Algorithm randomly permutes the list to generate the order in which candidates are interviewed.
- If this is done in such a way that each of the n! permutations is equally likely, then the previous analysis can be used.

Key point: No particular input elicits worst case behavior. (The algorithm performs badly only when the random permutation generated turns out to be unluckv".)

RANDOMIZED-HIRE-ASSISTANT

```
RANDOMIZED-HIRE-ASSISTANT(n)
randomly permute the list of candidates
best = 0            // candidate 0 is a least-qualified dummy candidate
for i = 1 to n
    interview candidate i
    if candidate i is better than candidate best
        best = i
        hire candidate i
```

Probabilistic analysis

- Probabilistic analysis is the use of probability to analyze problems.
- One important issue is what is the distribution of inputs to the problem.
- For instance , we could assume all orderings of candidates are equally likely.
- That is, we consider all functions rank: [0..n] --> [0..n] where rank[i] is supposed to be the *i*th candidate that we interview. So <rank(1), rank(2), …rank(n)> should be a permutation of of <1,…,n>
- There are n! many such permutations and we want each to be equally likely.
- If this is the case, the ranks form a **uniform random permutation**.

Indicator Random Variables

- In order to analyze the hiring problem we need a convenient way to convert between probabilities and expectations.
- We will use indicator random variables to help us do this.
- Given a sample space S and an event A. Then the **indicator random variable** I{A} associated with event A is define as:

$$I\{A\} = \begin{cases} 1 & \text{if A occurs ,} \\ 0 & \text{if A does not occur .} \end{cases}$$

Example

- Suppose our sample space S={H,T} with Pr{H}=Pr{T}=1/2.
- We can define an indicator random variable X_H associated with the coin coming up heads:

$$X_H = I\{H\} = \begin{cases} 1 & \text{if H occurs ,} \\ 0 & \text{if T occurs .} \end{cases}$$

- The expected number of heads in one coin flip is then

$$\begin{aligned} E[X_H] &= E[I\{H\}] \\ &= 1 \cdot Pr\{H\} + 0 \cdot Pr\{T\} \\ &= 1 \cdot (1/2) + 0 \cdot (1/2) \\ &= 1/2 \end{aligned}$$

Analysis of the Hiring Problem

- Let X_i be the indicator random variable which is 1 if candidate *i* is hired and 0 otherwise.
- Let
$$X = \sum_{i=1}^{n} X_i$$
- By our lemma $E[X_i] = Pr\{\text{candidate } i \text{ is hired}\}$
- Candidate *i* will be hired if *i* is better than each of candidates 1 through *i*-1.
- As each candidate arrives in random order, any one of the first candidate i is equally likely to be the best candidate so far. So $E[X_i] = 1/i$.

- Probability that the ith candidate is best of first i is 1 / i
    - Sanity Check: (Doing a few concrete examples as a sanity check is often a good idea)
        - i = 1, probability that the first candidate is the best so far = 1/1 = 1
        - i = 2: (1,2), (2,1) In one of the two permutations, 2nd candidate is the best so far
        - i = 3: (1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1) In two of the 6 permutations, the 3rd candidate is the best so far

- What is $E[X_i]$?
    - $E[X_i]$ = Probability that the ith candidate is hired
    - ith candidate hired when s/he is better than the i − 1 candidates that came before
    - Assuming that all permutations of candidates are equally likely, what is the probability that the ith candidate is the best of the first i candidates?
        - 1 / i

More analysis of hiring problem

$$E[X] = E[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} 1/i = \ln n + O(1)$$

Lemma Assume that the candidates are presented in random order, then algorithm Hire-Assistant has a hiring cost of $O(c_h \ln n)$

Proof. From before hiring cost is $O(m*c_h)$ where m is the number of candidate hired. From the lemma this is $O(\ln n)$.